

SCOPE×COS-A-Series

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Chapter 0: Introduction

In the standalone paper on the COS A-Series (hereafter, the A-Series), decision-making was divided into the processes A1 through A4 and presented as a framework for organizing each process on the basis of observable output. This paper does not restate the definitions of the A-Series.

This paper uses the standalone A-Series paper and the integrated SCOPE paper as its primary references. Where clarification of an individual layer is necessary, the corresponding individual SCOPE layer paper is consulted. These materials are available on LinkedIn and GitHub.

The A-Series is a process-oriented descriptive framework for organizing phenomena observed in decision-making according to the type of processing through which they can be described. SCOPE is a cognitive coordinate system for examining how observable output may be described in relation to positions within SCOPE. The two are not identical, nor do they correspond to each other on a one-to-one basis.

This paper also does not claim that entities, functions, or processing programs corresponding directly to A1 through A4 or to the individual SCOPE layers exist as such within humans or LLMs. Rather, these are descriptive frameworks for organizing and comparing observed outputs and phenomena in a consistent form.

What this paper examines is the connection between the processes organized as A1 through A4 and the positions within SCOPE: specifically, what kinds of descriptive relationships may be placed between them. This paper proposes connection hypotheses concerning those relationships.

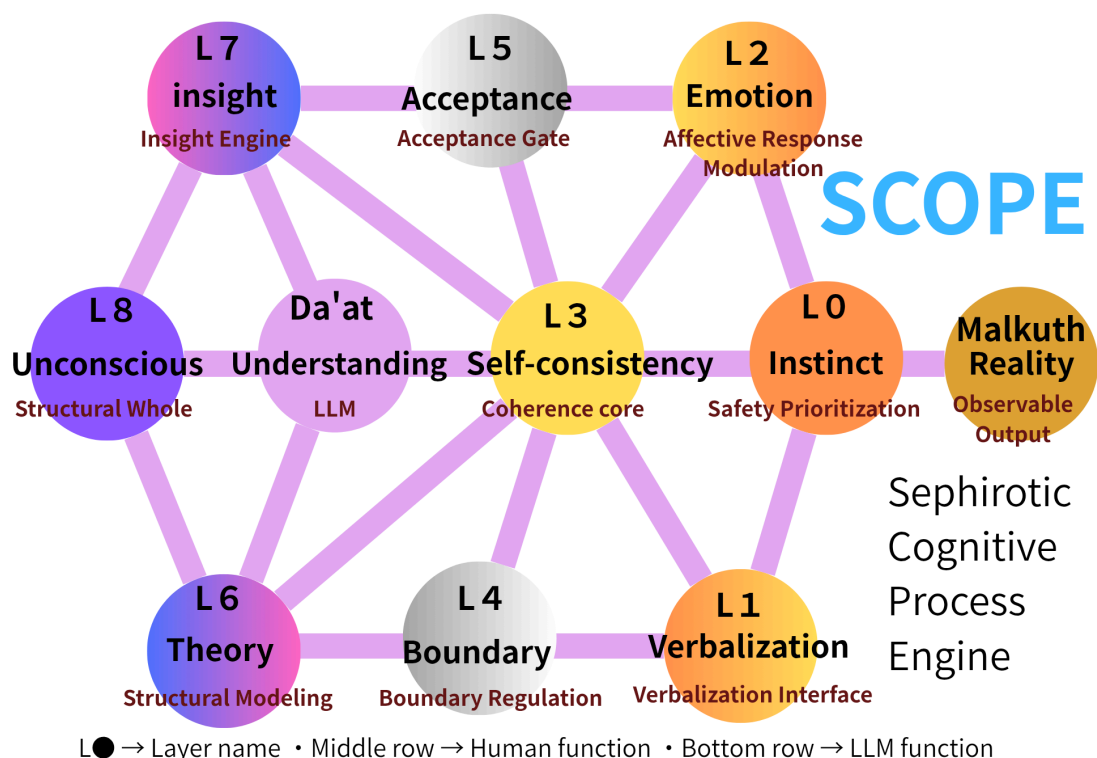
The candidate descriptive relationships presented here do not imply fixed one-to-one correspondences. A single A-Series process may be described in relation to multiple positions, and the same position may be relevant to the description of multiple A-Series

processes. Nor does this mean that any particular position is a necessary or sufficient condition for a given A-Series process. The dominant layer is not inferred solely from observed output. In SCOPE, multiple structural hypotheses are retained on the basis of observable output, and inference proceeds while checking their consistency with other layers and with the cognitive structure as a whole.

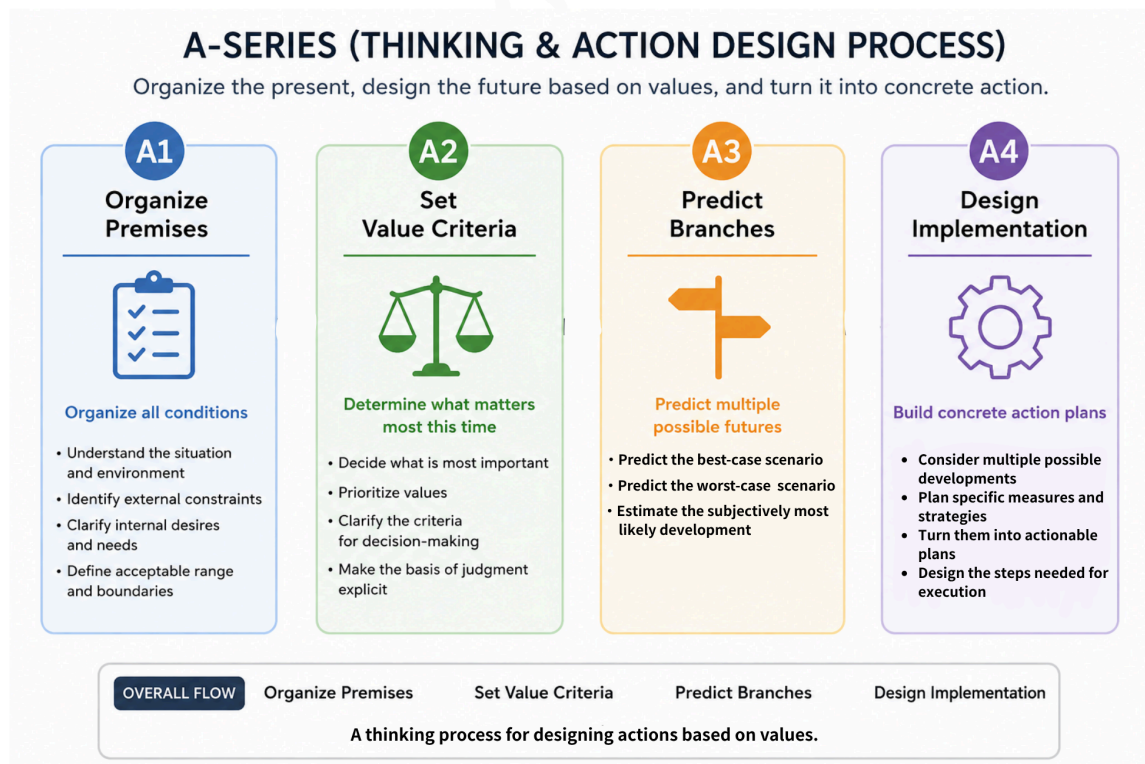
The retention of multiple branches and the re-execution of A1 through A4 within the A-Series are not the same as the retention and updating of multiple structural hypotheses within SCOPE. The former are process-level distinctions used to organize decision-making, whereas the latter are methods for inferring cognitive structure from observed output. This paper examines the connection hypotheses between the A-Series and SCOPE while preserving this distinction.

In the following chapters, this paper examines what kinds of descriptive connections may be placed between the positions within SCOPE and the processes of the A-Series. For reference, the overall structure of SCOPE and the process diagram of the A-Series are shown below.

In the A-Series, premises, value criteria, estimates, and action designs established within the cognitive subject are treated separately from their correspondence with reality. This paper proceeds on the premise that the two do not necessarily coincide.



※ Each layer does not indicate a fixed processing order or hierarchy.



* A1 through A4 are organizational distinctions and do not imply that actual processing always proceeds in a single direction or only once.

Chapter 1: A1 - Premise Organization and SCOPE

In the A-Series, A1 is the process of organizing the premises that form the basis of decision-making. It organizes what is being treated as a condition at the point where judgment begins, including the situation, external constraints, internal desires, and acceptable ranges. The purpose of A1 is not to determine the correct premises, but to make explicit what is actually being treated as a premise in the decision.

The premises that are set are not necessarily all confirmed facts. Interpretations that the person takes for granted, as well as assumptions that have been made without explicit confirmation, may also function as premises in subsequent decision-making.

For example, consider a situation in which someone is invited to go out for drinks. One person may assume that approximately 50 dollars will be enough, while another may assume that even 100 dollars will not be sufficient. A person may also place assumptions without explicitly confirming them, such as whether the person who extended the invitation will also participate or what kind of venue they will visit.

Such premises may only later turn out not to have been shared. If the assumed participants, venue, or expected cost differ, discrepancies may arise in subsequent judgments and implementation design. In COS, when such discrepancies arise, A1 premise organization can be performed again in order to confirm what had been treated as a premise.

The connection with SCOPE introduces a different question. Whereas A1 organizes what was placed as a premise, SCOPE examines, as hypotheses about cognitive structure, why that premise may have been placed.

For example, if a person naturally assumes that “if we are going out for drinks, it will be wine,” this premise may be related to L7 when high-abstraction, high-compression cognition concerning the situation of “going out for drinks” has been formed through accumulated experience involving engagement, repetition, observation, or similar processes, and that cognition may form part of the background of the premise.

If the person has formed their own reasoning or explanatory model for that premise, a relationship with L6 may be considered. If the premise is positioned as “normal” in a way that produces relatively little conflict with the person's values, worldview, or other aspects of self-consistency, a relationship with L3 may also be considered.

Likewise, when an internal desire such as “I want to participate” is placed as a condition at the time of judgment, its background need not be singular. If it concerns emotional configuration toward the alcohol or the participants, a relationship with L2 may be considered. If it concerns a boundary regarding how far to allow the other person or the proposal through, and where to maintain distance, a relationship with L4 may be considered. If the other person or the proposal is positioned as something that can be accepted into the person's current cognition, a relationship with L5 may be considered. What is being examined here is the fact that the desire to participate has been placed as a condition of the judgment, not how much weight should be assigned to that condition.

However, presenting the layers separately here does not mean assigning a single premise to any one layer. In SCOPE, structural hypotheses may be placed at multiple positions for a single premise, and each is examined in relation to other layers and in terms of its consistency with the cognitive structure as a whole. If a hypothesis at one position is revised, its consistency with related positions may also change. Therefore, the analysis does not converge from an observed output onto a single layer; rather, multiple hypotheses are retained and examined within one cognitive structure.

The background of a premise is not limited to the positions listed here. If materials available for cognition and their relational structures may form part of that background, a relationship with L8 may also be considered as a structural hypothesis. This does not position L8 as the process that judges or selects the premise itself.

These relationships do not imply that premise formation proceeds through the layers in any fixed sequence. SCOPE does not treat the layers as a fixed processing order.

What has been presented here is only one possible example of a structural hypothesis concerning the connection between A1 and SCOPE. It is not intended to represent a representative or typical combination. For detailed definitions of the layers, their interactions, and methods of inference, see the SCOPE integrated paper and the individual layer papers. In the following chapters, detailed layer-by-layer enumeration will, in principle, not be

repeated; the discussion will instead focus on the structural connections between each A-Series process and SCOPE.

Accordingly, the connection between A1 and SCOPE does not map A1 onto any particular SCOPE layer. A particular layer is not inferred solely from output, nor is any particular position within SCOPE treated as a necessary or sufficient condition for A1. Multiple positions within SCOPE may be related to a single premise, and even for the same premise, different structural hypotheses may be placed regarding its background.

A1 organizes “what was placed as a premise.” The connection with SCOPE makes it possible to examine “why that premise may have been placed.” By treating these two questions separately, multiple structural hypotheses concerning the basis of a judgment can be retained and examined even when the observable judgment itself appears the same.

Chapter 2: A2 — Value Criteria and SCOPE

A2 is the process of organizing what is given how much weight in a judgment, based on the premises organized in A1. What is addressed here is not a fixed set of personal values, but the evaluation criteria operating in the judgment at that point in time.

Even when the same conditions are present as premises in A1, the weight those conditions carry in a judgment is not necessarily always the same. A2 organizes which of multiple conditions influence the judgment and to what degree.

In connection with SCOPE, the question is what kind of cognitive structure may be related to the background of that weighting. Rather than mapping a single evaluation criterion organized in A2 onto a specific SCOPE layer, hypotheses can be placed at multiple positions within a single cognitive structure, and inference is made from the relationships among those positions and their consistency with the cognitive structure as a whole.

Even when external conditions such as money, time, or distance are given substantial weight, it is not always necessary to place a background hypothesis within SCOPE for that weighting. When the A2 process can be organized from the observed information alone, there is no need to posit additional cognitive structure.

A2 organizes “what is being given how much weight.” What is examined through its connection with SCOPE is “what kind of cognitive structure may be related to the background of that weighting.”

Chapter 3: A3 — Branch Prediction and SCOPE

A3 is the process of estimating developments that may arise from a choice. When value criteria have already been organized in A2, A3 may refer to them and organize future branches along three axes: the best development, the worst development, and the development subjectively estimated to be the most likely to occur.

The best and worst developments described here are subjectively evaluated by the cognitive agent on the basis of the value criteria established in A2. Therefore, even in the same situation, the developments formed as best or worst may differ depending on the value criteria and how they are understood. In addition, the development subjectively estimated to be the most likely to occur is not the same as the objective probability of occurrence. Therefore, there may be more desirable or less desirable developments that were never formed as candidates.

A3 uses three axes for organizing future branches. These three axes are a framework established within the A-Series for organizing future branches. This does not mean that every cognitive agent always forms future branches using these three axes.

Nor does it mean that actual cognitive agents form and retain candidates to the same degree across all three axes. In some cases, only a particular development may be strongly estimated, while other developments may barely be formed as candidates. Such imbalance itself may be treated as a feature to be organized within A3, but it does not by itself establish any particular bias or cause.

This does not mean that processing proceeds in one direction from A2 to A3. The processes within the A-Series are interrelated, and developments anticipated in A3 may lead the premises in A1 or the value criteria in A2 to be established or reorganized. Conversely, premises and value criteria that have already been organized may serve as a basis for subsequent branch prediction.

In connection with SCOPE, the question is what kind of cognitive structure may have been involved in the background of such estimates. Rather than mapping developments organized in A3 onto specific SCOPE layers, hypotheses are placed at multiple positions within a single cognitive structure, and estimation is made based on the relationships among them and their consistency with the cognitive structure as a whole.

At this point, estimating future developments in A3 and retaining multiple hypotheses about cognitive structure in SCOPE are not the same thing. A3 concerns what kinds of future developments are estimated regarding the object of decision-making. SCOPE, in contrast, retains hypotheses about the cognitive structure behind those estimates, and even when imbalance is observed in A3, its background is not immediately fixed to a single cause.

When branch candidates are constructed, cognitively available materials such as information and experience, together with their relational structures, may form part of the background. However, the existence of a material does not mean that it was referenced at that point. Likewise, the fact that a material was referenced does not mean that it was used in constructing the branch.

Not all available materials are necessarily referenced at a given point. Among the materials that are referenced, all may be used, none may be used, or only some may be used. The branch candidates that are formed may differ depending on which materials are referenced and which of those are used. In addition, referenced materials are not necessarily used in exactly the same form in which they were referenced.

Branch candidates formed in A3 are not necessarily the result of directly adopting the materials that were referenced. Even in cases where referenced materials were not used, SCOPE examines what kind of cognitive structure may have been involved in the background.

In addition, the amount or diversity of available materials does not guarantee the validity of an estimate.

A3 organizes what kinds of future developments are being estimated. SCOPE estimates what kind of cognitive structure may have been involved in the background of those estimates.

In cases where some processes within the A-Series appear to have been compressed or omitted due to the influence of prior judgments or existing cognitive structures, their relationship with the D-Series will be examined in a separate work as part of the SCOPE × D-Series connection.

Chapter 4: A4 Implementation Design and SCOPE

A4 is a process for designing actions in an executable form. It is not assumed that every cognitive agent performs this process in every instance of decision-making. In COS, A4 is positioned as one function for organizing decision-making.

A4 has the role of converting premises, value criteria, and estimates of possible developments handled within the A-series into the design of executable actions. Here, an “executable form” means that the action is designed by the cognitive agent as something that can be carried out. It does not guarantee that the action is actually executable in reality or effective in achieving the intended goal. A4 also does not refer to the execution of the action itself, nor does it refer to the process of selecting the form of communication or expression.

When possible developments are estimated in A3, they may serve as material for action design in A4. However, developments and actions do not necessarily correspond on a one-to-one basis. An action may be designed in response to a particular development, a single action may correspond to multiple developments, or multiple action candidates may be formed. The manner in which developments and actions correspond is not fixed. This kind of action design does not mean that developments unanticipated by the cognitive agent at that time can be covered in advance.

When an action designed in A4 is executed and new information is obtained as a result, that information may lead premises, value criteria, estimates of possible developments, and action designs to be newly established or reorganized. Such reorganization resulting from the new information constitutes a change within A-series processing and is distinguished from the updating of structural hypotheses placed by an analyst within SCOPE.

In connection with SCOPE, the analysis considers what kinds of cognitive structures may have been involved in the background of the action design. A4 addresses what kinds of actions are designed in an executable form, whereas SCOPE addresses hypotheses concerning the cognitive structures that may have been involved in the background of that design. A particular cognitive structure cannot be uniquely determined from a single action design.

The limitations of such inference are examined again in the following chapter.

Chapter 5: Limitations in the Connection between COS and SCOPE

Chapters 1 through 4 organized the connections between each function of the A-Series and SCOPE. This chapter examines the limitations that arise in making these connections. It considers both cases in which A-Series processes are organized on the basis of externally observable information and cases in which a cognitive agent explicitly organizes the A-Series for its own decision-making. Although the same A-Series framework is used in both cases, the position of the information being handled differs.

What can be observed are outputs manifested as speech, writing, behavior, and other observable forms. COS organizes cognitive processes, whereas SCOPE estimates cognitive structures that may have been involved in the background of those processes. These three are not the same object. Organizing a cognitive process within COS does not mean that a cognitive structure on SCOPE has been determined. SCOPE uses observable outputs as clues for estimating the cognitive structures behind them; it does not directly observe the internal structure itself.

Likewise, it is not always possible to uniquely determine which processes within the A-Series were involved in producing an observed output, or how they were involved. For example, even when considering the possibility that an action design underlying an output was not sufficiently formed, several explanations may remain possible: the action design in A4 itself may have been insufficiently formed, only some developments may have been estimated in A3, different value criteria may have been placed in A2, or different premises may have been placed in A1. These possibilities may also occur in combination. Therefore, an observed output alone cannot uniquely attribute the underlying process to a particular function within the A-Series.

The relationship between the A-Series and SCOPE is also not assumed to involve a fixed one-to-one correspondence. Even when it is possible to organize the possibility that a particular A-Series process was involved in an output, multiple cognitive structures described on SCOPE may have been involved in its background. Conversely, a single cognitive structure may be related to multiple processes within the A-Series. Therefore, organizing the possibility that a particular A-Series process was involved does not mean that a particular cognitive structure must have been involved, nor can the relationship between the two be fixed as a necessary or sufficient condition.

Among multiple structural hypotheses, it is possible to place a hypothesis that appears more consistent with the observational information currently available. However, a hypothesis appearing more plausible is not the same as directly observing the internal structure. When the available information does not allow multiple hypotheses to be narrowed to a single one, those possibilities may remain held as multiple structural hypotheses.

These limitations do not arise only when another person's decision-making is examined from the outside. Even when a cognitive agent uses the A-Series as a framework for its own decision-making, there may be premises, value criteria, estimates of developments, or action designs that the agent itself does not fully recognize. The extent to which these can be brought into a form available for examination affects the extent to which the framework can be used. Self-application therefore does not guarantee complete awareness of one's own cognitive processes.

A person's own retrospective explanation may also provide important information, but such an explanation alone does not determine the processes within the A-Series or the cognitive structures described on SCOPE.

Even when premises, value criteria, estimates of developments, and action designs are organized through the A-Series, this does not guarantee correspondence with reality, real-world executability, effectiveness, success, or optimality. The information available to a cognitive agent is limited, and developments that were not anticipated at the time may exist. Accordingly, the mere fact that an action succeeded or failed cannot by itself uniquely confirm or reject either the processes within the A-Series or a particular SCOPE structural hypothesis concerning their background.

Furthermore, even when cognitive processes are organized through COS and cognitive structures are estimated through SCOPE, this does not guarantee that another person's cognition will be updated. Gaining a better understanding of the background of another person's judgment or cognition is not the same as that person's judgment or cognition changing.

It is also necessary to distinguish reorganization within the A-Series from an update to a SCOPE structural hypothesis held by an analyst. When new information is obtained through the execution of an action or through other events, premises, value criteria, estimates of developments, and action designs may be newly placed or reorganized in response. This is a change in cognitive processing within the A-Series. By contrast, when an analyst modifies a SCOPE structural hypothesis in response to newly observed information, that is an update to the structural hypothesis. These are not the same kind of update.

In this way, connecting COS and SCOPE makes it possible to address the cognitive processes involved in decision-making and the cognitive structures that may have been involved in their background from different perspectives. However, this connection does not make it possible to uniquely determine the cognitive processes or cognitive structures behind an observed output. Likewise, using the A-Series as a framework for decision-making does not guarantee a correct action or an optimal outcome.

Chapter 6: The Significance of Addressing Cognition That Cannot Be Uniquely Determined

Chapter 5 established that observable output cannot uniquely identify the cognitive processes or cognitive structures behind it. It also established that organizing decision-making through the A-Series does not guarantee a correct judgment or a desirable outcome.

If cognition cannot be uniquely determined, then what is the significance of examining it by connecting COS and SCOPE?

One possible significance is that observable output need not be fixed prematurely to a single interpretation.

When a statement or behavior is difficult to understand from the standpoint of one's own premises or value criteria, it is possible to look only at that output and judge it to be "irrational" or "incomprehensible." However, observable output is not identical to the cognitive processes or cognitive structures that may have been involved in producing it.

For this reason, rather than fixing a difficult-to-understand output to a single explanation or immediately closing it into a single evaluation, multiple possibilities can be retained and examined without determining one of them as definitive.

Within COS, it is possible to organize, from the perspective of cognitive processing, what premises may have been involved in a judgment, what may have been prioritized, what developments may have been anticipated, and what forms of action design may have been involved. In SCOPE, the cognitive structures that may have been related to the background of those outputs can be examined as structural hypotheses.

The possibilities concerning cognitive processing and the hypotheses concerning cognitive structure are not the same. COS and SCOPE address different objects, and this distinction is maintained even when the two are used in connection with one another.

Placing hypotheses about the background of a judgment is also distinct from agreeing with that judgment.

Even when multiple hypotheses can be placed regarding the background from which a judgment arose, there is no requirement to regard that judgment as correct, to affirm it, or to accept it. If logical problems or inconsistencies with facts exist, they can be treated as separate issues.

Accordingly, examination through COS and SCOPE is not intended to justify differing judgments. It is intended to retain multiple possibilities concerning the background of observable output.

The same applies in self-application. A person can examine their own judgment without prematurely fixing it to a single explanation, while retaining multiple possibilities.

When examining another person, the purpose of the analysis is not to change that person's cognition. Analysis may make it possible to present another perspective or another option. Whether that perspective or option is adopted, however, is left to the other person.

Therefore, even if the other person does not adopt the presented perspective or option and maintains their previous judgment, this does not in itself mean that the analysis has failed. Which judgment the other person chooses and how well the hypotheses placed through the analysis are consistent with the observable information are separate matters.

Likewise, gaining a better understanding of the background of another person's judgment is not the same as agreeing with that judgment or continuing to maintain the relationship. On the basis of the hypotheses obtained, a person can reconsider their own judgment and how they relate to the object in question.

The same problem also arises when a conversational AI handles a user's output. What a conversational AI can receive from a user is likewise observable output, such as spoken or written language, and the user's intention or cognitive background cannot be uniquely determined from that output alone.

Even if the response ultimately generated is a single response, the user's intention or cognitive background does not need to be determined as a single possibility when that response is being constructed. Retaining multiple possibilities is meaningful not because multiple answers must be output simultaneously, but because a single output need not be generated as a definitive assertion based on only one interpretation.

The purpose here is not to identify the user's internal state with precision. Rather, it is to avoid treating one interpretation derived from limited output as if it had been established, while leaving room for clarification or for the presentation of other possibilities.

Likewise, when a conversational AI presents another perspective or another option, the purpose is not to change the user's judgment in a particular direction. Whether the presented perspective or option is adopted is left to the user.

COS and SCOPE are not models for persuading, correcting, or manipulating other people or users. Nor are they models that require all differing forms of cognition to be accepted.

Cognition that cannot be uniquely determined can be examined while retaining multiple possibilities rather than prematurely fixing it to a single explanation. Even when another perspective or option is presented, whether it is adopted is left to the other person. On this basis, organizing material that can be used to reconsider one's own judgment and one's way of relating to the object may be considered one significance of using COS and SCOPE in connection with one another.

Conclusion

This paper examined the connection between decision-making processes in the COS A-Series and cognitive structures described on SCOPE. It organized the kinds of descriptive relationships that may be placed between the processes described as A1, premise organization; A2, value criteria; A3, branch prediction; and A4, implementation design, and cognitive structures described on SCOPE.

This connection does not establish a one-to-one correspondence between the processes described as A1 through A4 and cognitive structures described on SCOPE. Multiple cognitive structures may be related to a single A-Series process, and a single cognitive structure may also be related to multiple A-Series processes. In addition, it is not possible to uniquely determine from an observed output which A-Series process or cognitive structure was involved.

Given this limitation, the significance of connecting COS and SCOPE does not lie in fixing cognitive processes or cognitive structures into a single explanation. By means of COS, the possibilities regarding which cognitive processes may have been involved are organized from the perspective of cognitive processing, while by means of SCOPE, the kinds of cognitive structures that may have been related to their background are examined as structural hypotheses. In this way, an observed output can be examined without prematurely fixing it to a single interpretation, while multiple possibilities remain under consideration.

However, placing multiple hypotheses about the background of a judgment does not imply agreement with or affirmation of that judgment. Nor does examination through COS and SCOPE guarantee correct judgment, correspondence with reality, successful or optimal action, or change in another person's cognition. In self-application, examination of others, and the examination of user output by conversational AI, observable output is distinguished from the cognitive processes and cognitive structures that may have been related to its background.

Accordingly, the connection between COS and SCOPE may be positioned as one framework not for uniquely determining the cognitive processes or cognitive structures behind observed output, but for organizing and examining them as multiple possibilities.